32-bit Microcontrollers

HC32L072 / HC32L073 / HC32F072 Series

Hardware Development Guide

适用对象

系列	产品型号
HC32L072	HC32L072PATA HC32L072KATA HC32L072JATA
HC32L073	HC32L073PATA HC32L073KATA HC32L073JATA
HC32F072	HC32F072PATA HC32F072KATA HC32F072JATA

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1 Summary

This application note mainly introduces the basic hardware design and precautions of the HC32L072 / HC32L073 / HC32F072 series chips, including the minimum system, common interfaces, common peripherals, and chip packaging.

This application note includes:

- Minimum system design points
- Common interface precautions
- Common peripheral reference design
- Chip packaging introduction
- Development board request

Note:

- This application note is an application supplement for the HC32L072 / HC32L073 / HC32F072 series and cannot replace the user manual. Please refer to the user manual for specific functions and register operations and other related matters.

2. Function Introduction

This article mainly introduces how to use the HC32L072 / HC32L073 / HC32F072 series chips for hardware system design. From the perspectives of the minimum system, common interfaces, and common peripherals, it elaborates on how to design this series of chips in actual product applications and how to make product design more stable and reliable.

3. Minimum System Design Points

When designing the minimum system, hardware engineers need to pay attention to the following points: power supply processing, reset processing, clock processing, debug interface processing, port processing, etc.

3.1 Power supply processing

Each (DVCC/AVCC) pin of the chip requires a decoupling capacitor. When laying out, the capacitor is close to the corresponding power pin, as shown in Figure 1.

The VCAP pin of the chip is the output of the core power supply LDO, which is limited to internal circuit use and is strictly prohibited from connecting any load externally; two capacitors 1uF+100nF are required, as shown in Figure 1.

Each power pin (DVCC/AVCC DVSS/AVSS) of the chip must be assigned a reasonable power supply voltage to ensure that the chip is in normal working condition.

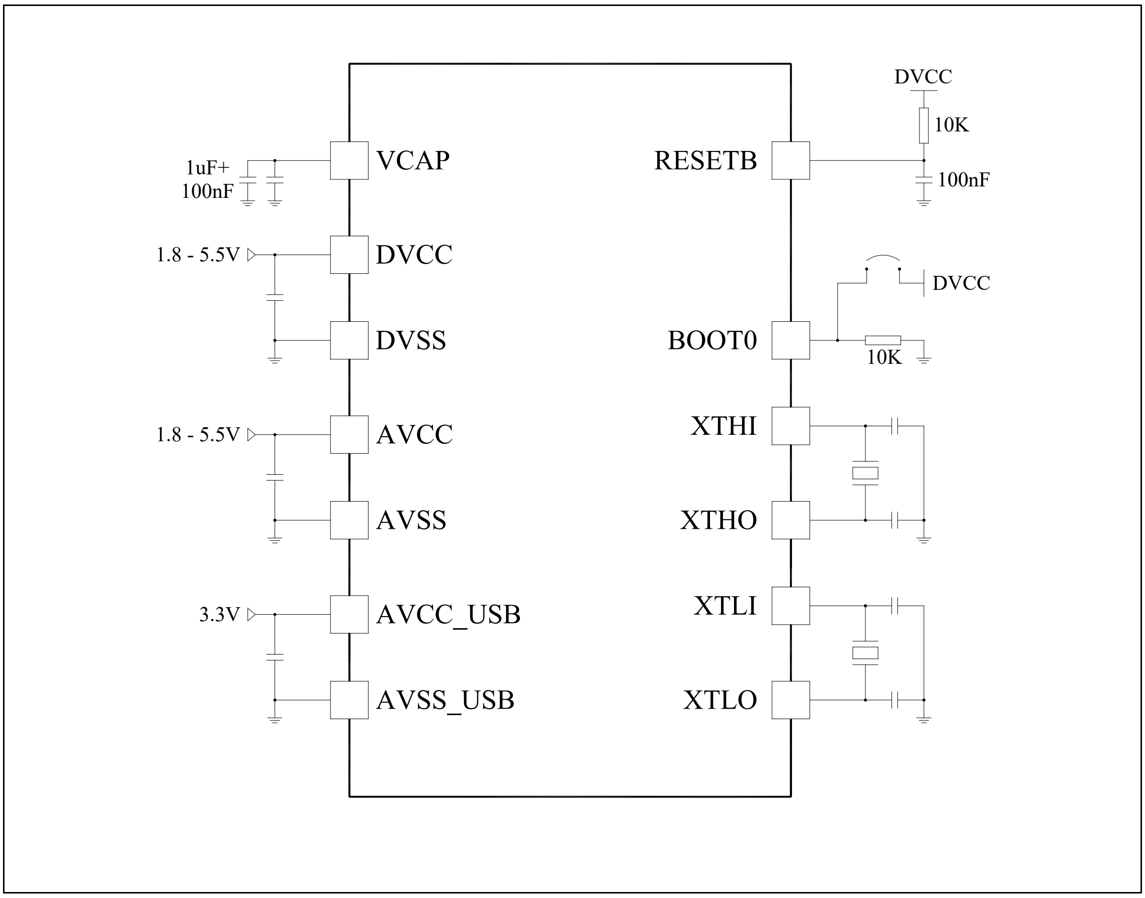

Figure 1 Minimum system power supply block diagram

Note:

- *It is strictly forbidden to connect any load outside VCAP.*

- *The power supply voltage range is: 1.8V≤DVCC/AVCC ≤5.5V*

3.2 Reset circuit

When designing, please connect a capacitor and a resistor between the chip's reset (RESETB) pin and ground (DVSS) to form an RC circuit, as shown in Figure 2.

Usually, when applied, a touch button is connected in parallel to achieve the purpose of manual reset, as shown in Figure 2.

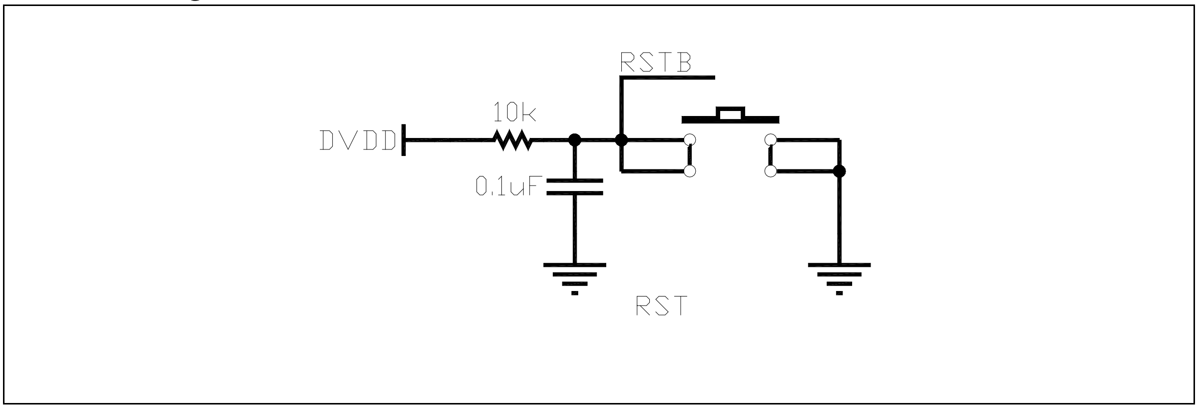

Figure 2 Common reset circuit

Note:

- *Avoid interference with the reset signal.*

3.3 Clock design

This series of chips has two external clock sources, the high-speed clock is not higher than 32MHz, and the low-speed clock source is 32.768KHz. When designing, please be sure to confirm the detailed parameters of the crystal (frequency, packaging, accuracy, etc.) and select reasonable matching capacitors, as shown in Figure 3. In the application, the resonator and load capacitors must be close to the pins of the oscillator to reduce output distortion, startup stabilization time and noise radiation; when laying out, please make an isolation ring around the crystal, lay the ground sufficiently, and control the clock noise within the isolation ring, as shown in Figure 4.

Note:

- It is strictly forbidden to place the filter capacitor of the VCAP pin close to the crystal; avoid running signal lines under the crystal.

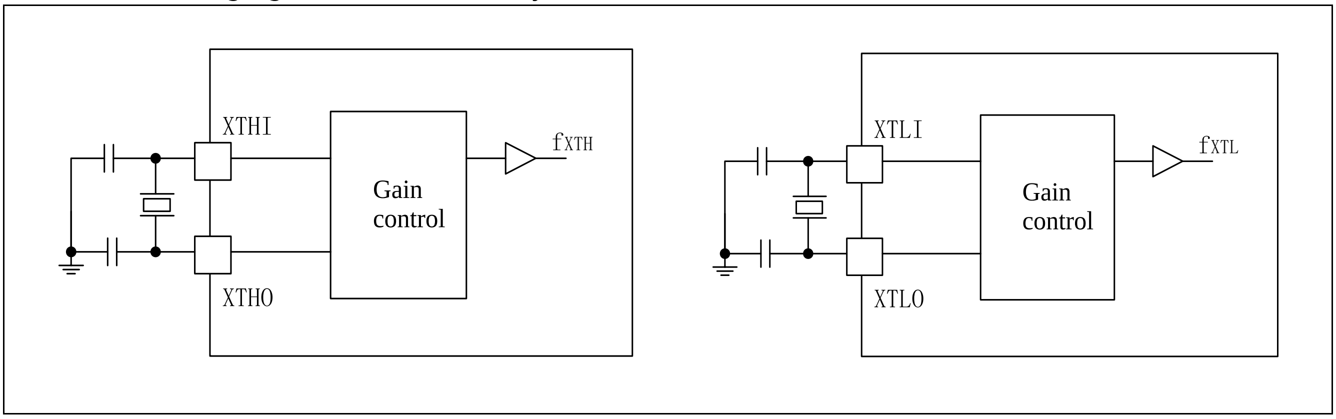

Figure 3 Crystal peripheral circuit

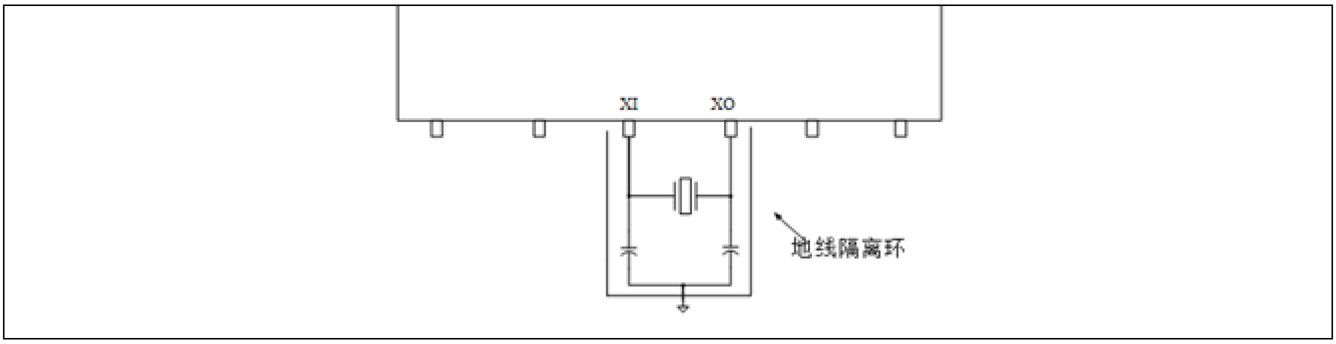

Figure 4 Crystal isolation ring processing

3.4 Debug interface

The chip debugging interface is used for code debugging, burning, etc., as shown in Figure 5.

This series of chips uses the ARM Cortex-M0+ core, which has a hardware debugging module SWD that supports complex debugging operations.

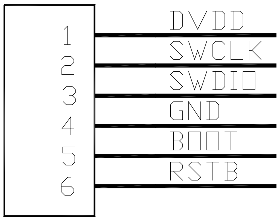

Figure 5 SWD interface

3.5 Port processing

In addition to the above-mentioned function pins, general IO ports do not require special processing methods, and unused input ports can be directly floated.

Note:

- When using digital-to-analog conversion, the Vref pin needs to be connected to an external capacitor to ensure the accuracy of the reference source.

4 Common interface considerations

This chapter briefly introduces some common interface hardware design considerations.

4.1 UART interface

UART interface design, TX/RX signal line connected to pull-up resistor, the resistance value cannot be too small.

Note:

- *The pull-up resistors of both communicating parties cannot be repeated.*

4.2 SPI interface

SPI interface design, in principle, all pull-ups, the resistance value cannot be too small.

Note:

- *The pull-up resistors on the bus cannot be repeated.*

4.3 I2C interface

I2C interface design, signal line connected to pull-up resistor, the resistance value cannot be too small.

Note:

- *The pull-up resistors on the bus cannot be repeated, the pull-up resistor value of this series of chips is recommended to be 1K.*

4.4 USB interface

The USB full-speed (USBFS) controller provides a set of USB communication solutions for portable devices. The USBFS controller supports device mode, and the chip integrates full-speed PHY. Full-speed (FS, 12Mb/s) transceiver is supported in device mode. The USBFS controller supports all four transfer modes defined by the USB 2.0 protocol (control transfer, bulk transfer, interrupt transfer and synchronous transfer).

Note:

- *The DP DM signal lines of USB must be treated with equal length differential and 90 ohm impedance to ensure excellent performance.*

5 Common peripheral reference design

5.1 LCD screen circuit

The MCU chip contains a built-in LCD driver circuit that can directly drive the LCD screen on the board. Usually, the bias voltage of the LCD screen has three sources: internal resistor voltage division, external resistor voltage division, and external capacitor voltage division. In order to reduce power consumption, the hardware is configured to external capacitor voltage division mode by default. The LCD selects 4-word 8-segment code display, working voltage 3V3, and mode is semi-transparent without backlight. Due to the built-in driver circuit, the LCD screen connection method is simple. Just connect COM and SEG to the corresponding function pins of the MCU, as shown in the figure below:

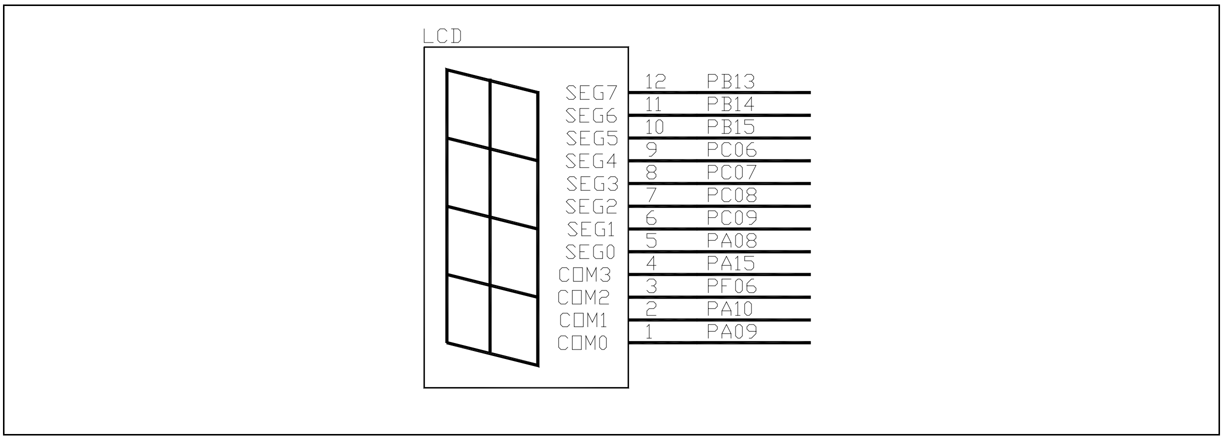

Figure 6 LCD screen signal connection diagram

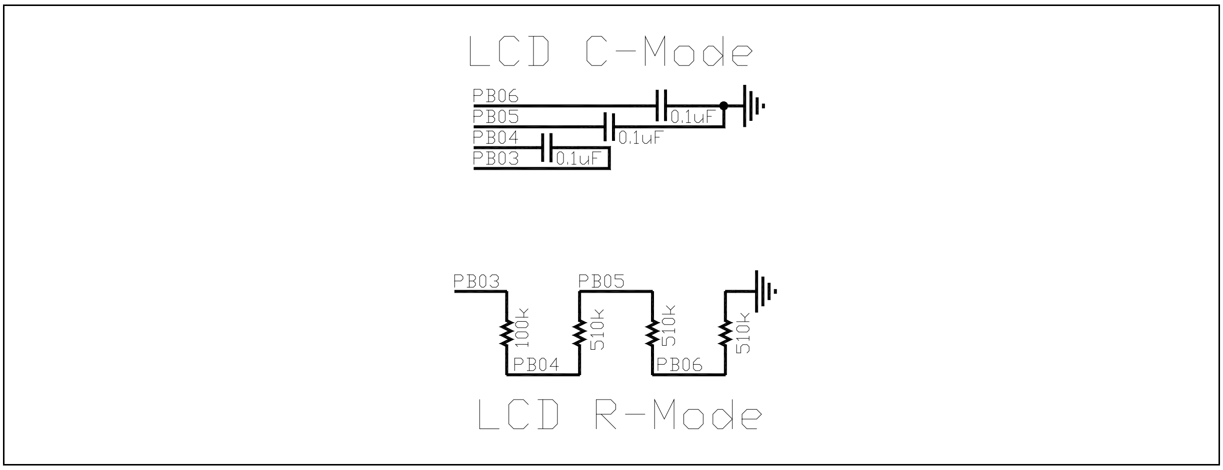

Figure 7 Bias external voltage divider mode

5.2 CAN transceiver

CAN (Controller Area Network) bus is a bus standard that enables microprocessors or devices to communicate with each other without a host. This module complies with CAN bus protocol 2.0A and 2.0B and is upwardly compatible with CAN-FD. The CAN bus controller can process data transmission and reception on the bus. In this product, CAN has 8 groups of filters. NXP's TJA1042 is recommended, and the reference design is shown in the figure below.

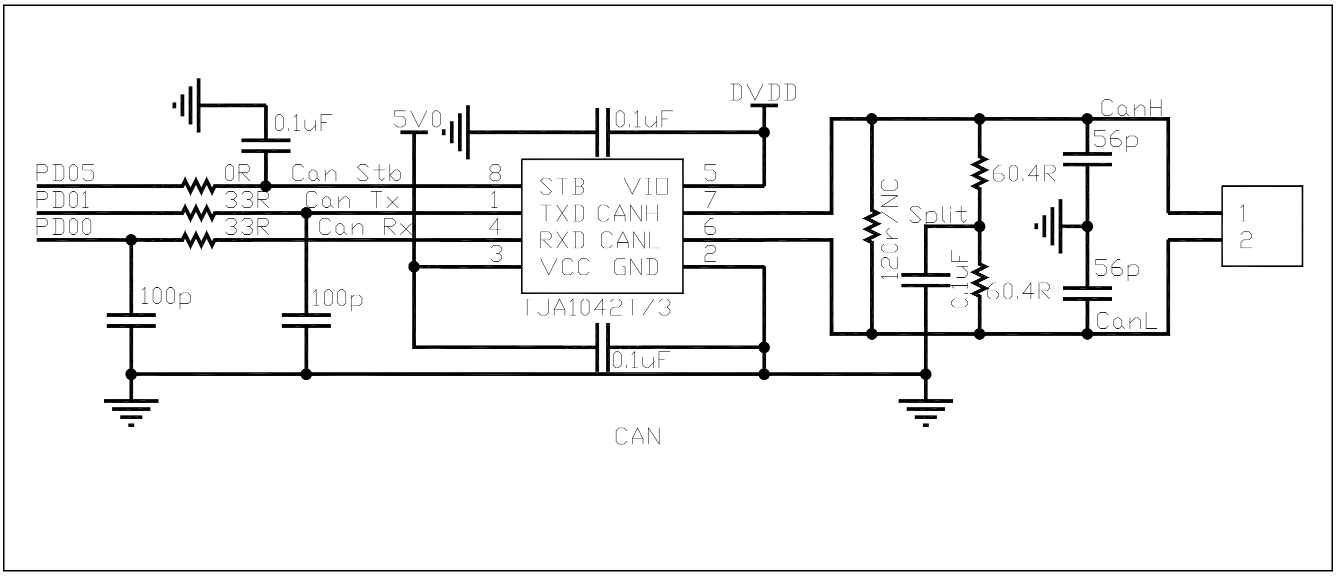

Figure 8 CAN reference design

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5.3 I2C memory

I2C is a two-wire bidirectional serial bus that provides a simple and efficient method for data exchange between devices. The I2C standard is a true multi-host bus with conflict detection and arbitration mechanisms. It can prevent data conflicts when two or more hosts request control of the bus at the same time.

The I2C bus controller can meet various specifications of the I2C bus and support all transmission modes for communicating with the I2C bus.

The I2C bus uses "SCL" (serial clock bus) and "SDA" (serial data bus) to connect devices to transmit information.

For I2C memory, Belling's BL24C series is recommended, and the reference design is shown in the figure below.

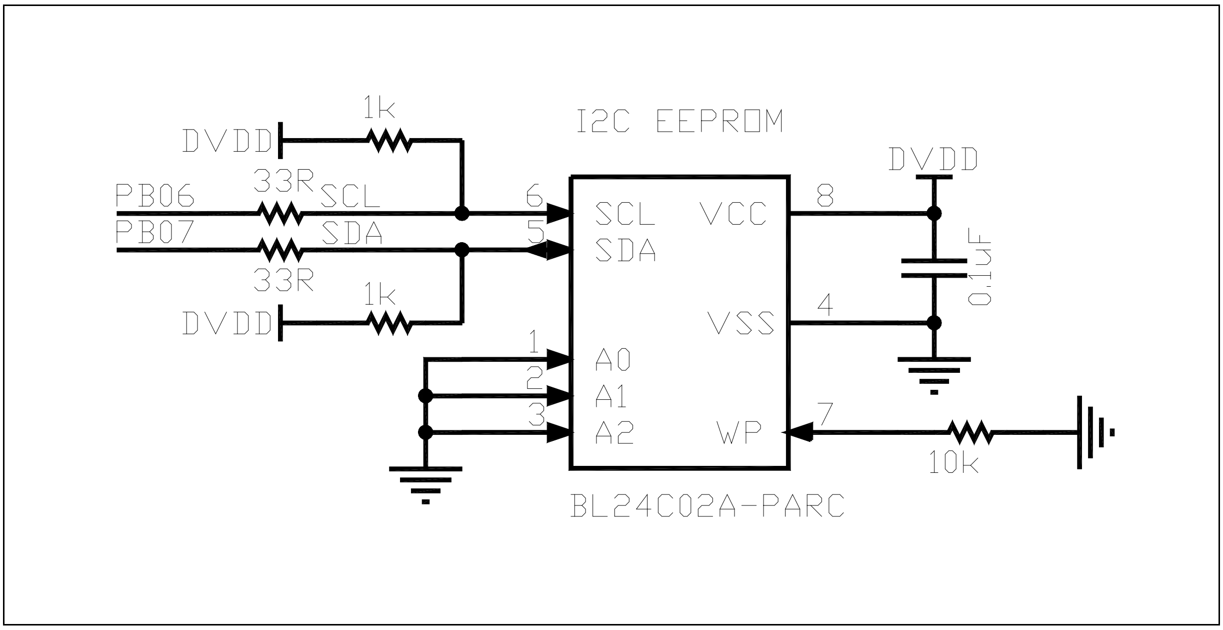

Figure 9 EEPROM reference design

5.4 SPI memory

The SPI (Serial Peripheral Interface) bus is a synchronous serial peripheral interface that enables the MCU to exchange information with various peripheral devices in a serial manner.

The SPI interface uses 4 lines: serial clock (SCK), master output/slave input line (MOSI), master input/slave output line (MISO), and low-level active slave select line (SSN).

For SPI memory, Cypress's S25FL series is recommended, and the reference design is shown in the figure below.

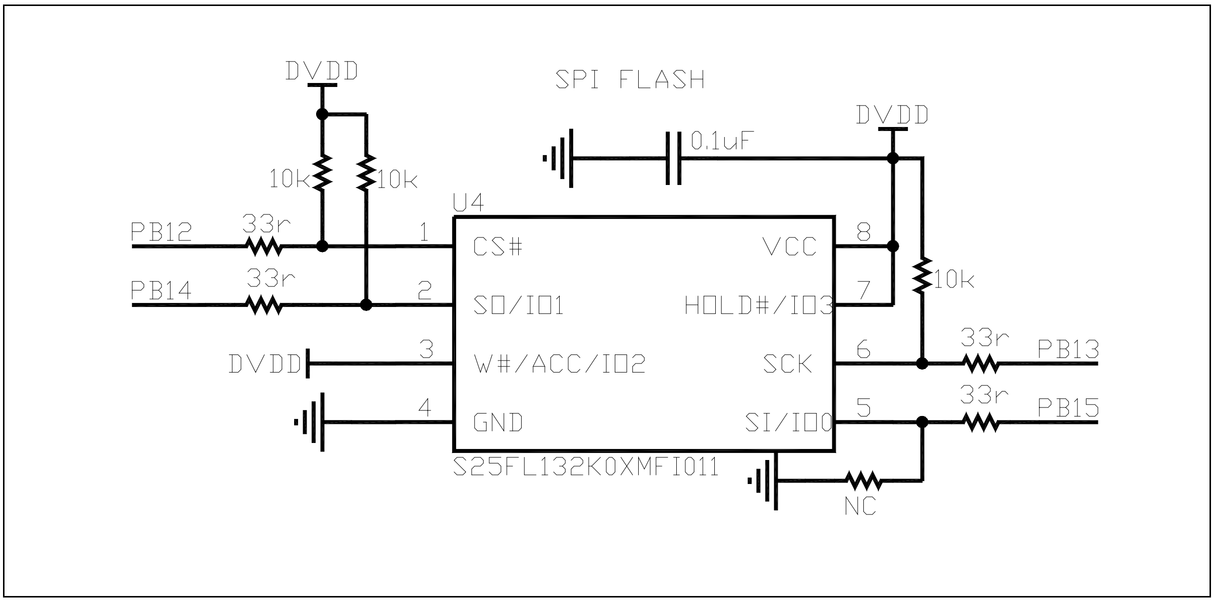

Figure 10 SPI FLASH reference design

5.5 I2S codec

I2S (Inter-IC Sound) bus, also known as the integrated circuit built-in audio bus, is a bus standard developed by Philips for audio data transmission between digital audio devices. This bus is specifically used for data transmission between audio devices and is widely used in various multimedia systems.

For I2S audio codec, the WM8731 series is recommended, and the reference design is shown in the figure below.

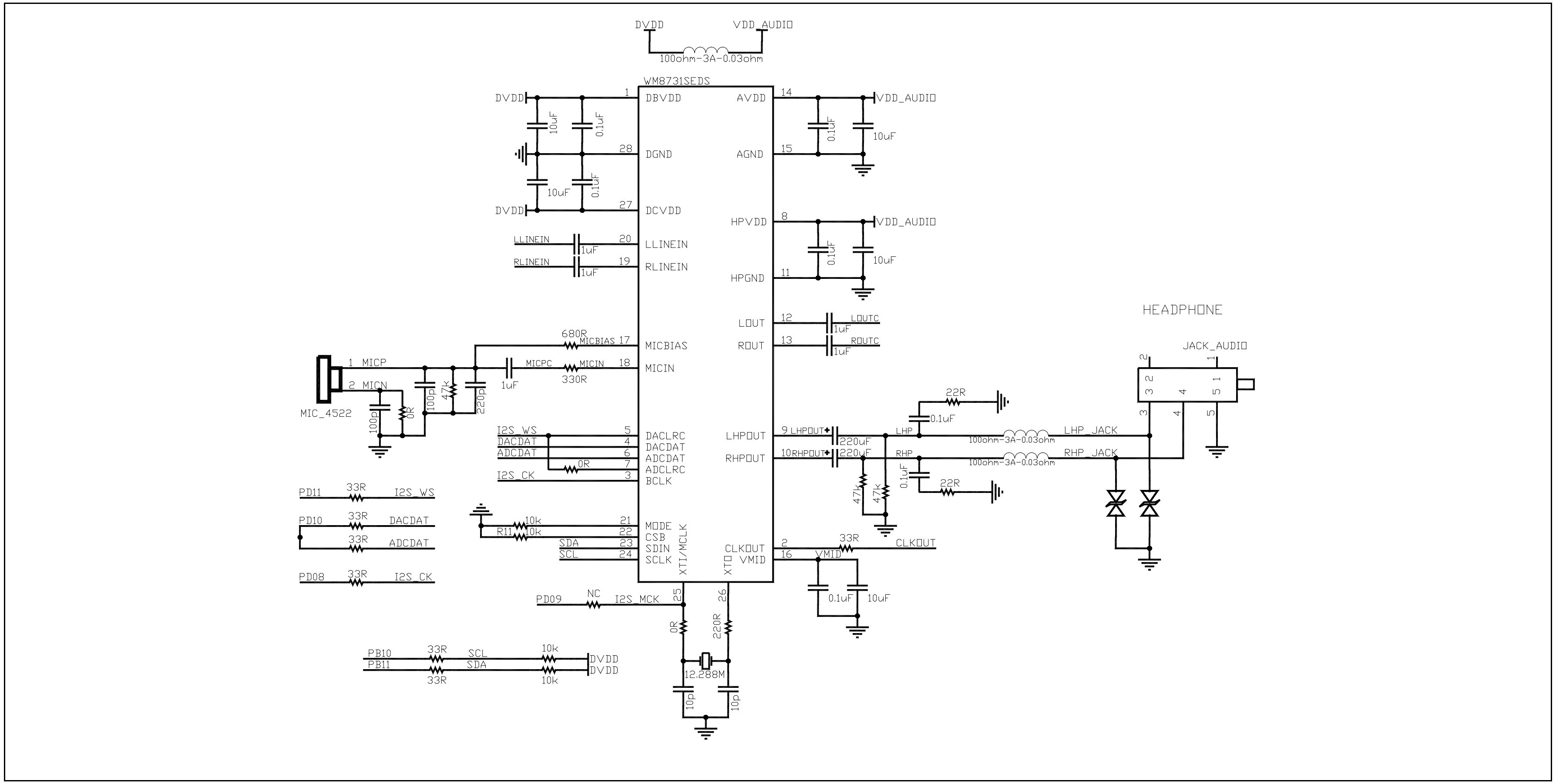

Figure 11 CODEC reference design

6 Chip packaging introduction

Our company provides all PCB package libraries of this series of chips for free.

For details, please refer to: http://www.xhsc.com.cn/list/73/

7 Sample and development board application information

Our company provides samples and development boards of this series of chips.

To apply for samples or purchase development boards, please refer to: http://www.xhsc.com.cn/list/73

8 Other information

Technical support information: http://www.xhsc.com.cn

9 Version information & contact information

Date	Version	Content
2019/7/26	Rev1.0	First version released.
2022/7/15	Rev1.1	Company Logo updated.

If you have any comments or suggestions during the purchase and use process, please feel free to contact us.

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